

Ep08 Optical Activity

OBJECTIVE

1. To study the principle of the reading of the angular vernier.
2. To learn how to measure the concentration of general sugar solutions using optical activity.

THEORY

Certain materials and solutions are able to rotate the plane of polarization of the light passing through them. This phenomenon is called optical activity.

Optical activity occurs in certain crystals (e.g., quartz) when plane polarized incident light is incident along certain directions with respect to symmetry axes associated with the particular crystal structure. Some liquids (e.g., turpentine) and solutions (e.g., sugar solutions) also exhibit optical activity. The extent of the rotation of the plane of polarization depends on the distance l traveled through the crystal or liquid and the concentration C of the solutions. In solutions, the angle of rotation ϕ of the plane of polarization is written empirically as

$$\phi = \alpha \frac{C \cdot l}{100} \quad (1)$$

where α is a proportionality constant known as the **specific optical rotating power** and is a function of both wavelength and temperature. The dependence on the wavelength of the light leads to rotary dispersion, analogous to the variation of the index of refraction with wavelength.

When the temperature is $t^\circ\text{C}$ and the incident light is sodium light, the specific optical rotating power of general sugar solutions is expressed:

$$\alpha = 66.5^\circ - 0.0184(t - 20^\circ) \quad (2)$$

substituting (2) into (1), we obtain:

$$C = \frac{100\phi}{l} \cdot \frac{1}{66.5^\circ - 0.0184(t - 20^\circ)} \quad (3)$$

if we neglect the effect of the temperature, then:

$$C = \frac{100 \times \phi}{66.5^\circ \times l} \quad \text{或} \quad C = 1.503 \frac{\phi}{l} \quad (4)$$

So, if the distance l traveled through the sugar solutions and the angle of rotation ϕ of the plane of polarization can be measured, we can calculate the concentration of the sugar solutions from equation (4).

Note: l is the distance traveled through the sugar solutions in **decimeter**.

Optical activity is used in many of the sciences including geology(crystalline minerals), chemistry(concentrations of solutions, isometric structures of molecules), biology(biologically important molecules with spiral structures such as DNA and RAN), structural engineering(for optical stress analysis), electrical engineering(liquid crystals), and physics for a wide variety of purpose and application.

PROCEDURE

(1) Determine the zero readings of the analyzer plate. Put the tube full of water into the device used to measure the concentration of liquids. Turn on the light. After a few minutes, observe from the telescope, you can see the boundary between brightness and darkness in the field of vision. Rotate the analyzer until the three parts in the vision field are equally dark, and then record the readings in dial of the device. Repeat the procedure three times; the average of the readings is the zero number.

(2) Measure the concentration C and purity P of the sugar solutions. Put the tube full of sugar solution into the device used to measure the concentration of liquids. Rotate the analyzer until the three parts in the vision field are equally dark too, and then record the readings in dial of the device. Repeat the procedure three times; calculate the average of the readings, the difference between the average and the zero number is the angle of rotation ϕ of the plane of polarization.

(3) Record the length of the tube l and the standard concentration given by the lab, then we can calculate the concentration C and the purity P .

DATA RECORDING AND PROCESSING

Items times	ϕ_0		ϕ	
	left	right	left	right
1				
2				
3				
Average				

$$\phi = \frac{(\bar{\phi}_{left} - \bar{\phi}_{0left}) + (\bar{\phi}_{right} - \bar{\phi}_{0right})}{2}$$

=

C=

P=

QUESTION

Why is the tube full of water put into the device rather than nothing in the device, when we determine the zero readings of the analyzer plate?